

Customer Churn Prediction

Using AI-ML and Deep Learning

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https://github.com/shreeya9094/Customer_Churn_Prediction_E2E_AI_ML_Project

<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Presentation Overview

- Objective
- Problem Statement
- Dataset & Methodology
- Model Architecture
- Results & Evaluation
- Business Insights
- Future Improvements
- Conclusion

Objective

- Develop a Machine Learning and Deep Learning based model to predict telecom customer churn accurately.
- Analyze customer behavior and identify key factors influencing customer attrition.
- Help businesses improve customer retention strategies and reduce revenue loss through data-driven insights.
- Compare the performance of multiple Machine Learning and Deep Learning algorithms using evaluation metrics.
- Enable organizations to take proactive actions for customer satisfaction and long-term business growth.

Problem Statement

- Customer churn directly impacts telecom company revenue.
- Acquiring new customers is more expensive than retaining existing customers.
- Need an intelligent predictive system to identify high-risk customers early.

Dataset Information

- IBM Telco Customer Churn Dataset from Kaggle.
- Contains demographic, service usage, financial, and tenure information.
- Target Variable: Churn (Yes / No).
- Problem Type: Binary Classification.

Approach & Methodology

- Performed data preprocessing and cleaning.
- Converted Total Charges to numeric values.
- Handled missing values using median imputation.
- Applied One-Hot Encoding for categorical variables.
- Scaled numerical features using Standard Scaler.
- Split dataset into training and testing sets.

Models Implemented

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier
- Artificial Neural Network (ANN)

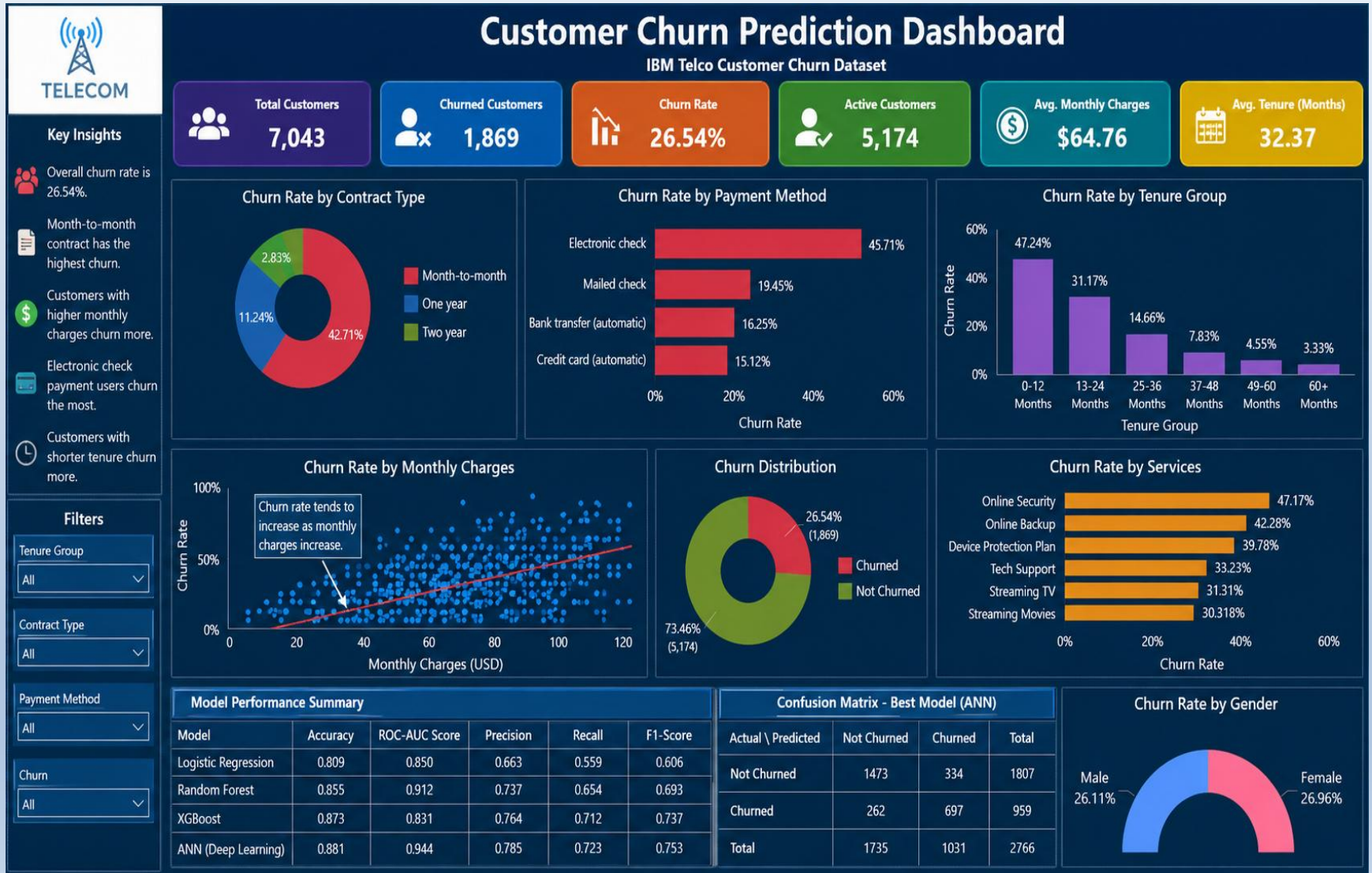
Deep Learning Architecture

- Input layer with customer features.
- Three hidden layers using ReLU activation.
- Dropout layers used to prevent overfitting.
- Sigmoid activation in output layer.
- Binary Cross entropy loss function with Adam optimizer.

Results & Evaluation

- Evaluation Metrics Used:
 - Accuracy Score
 - ROC-AUC Score
 - Confusion Matrix
 - Classification Report
- ML and DL models successfully predicted churn patterns.

Customer Churn Prediction Dashboard



Key Business Insights

- Month-to-month contract customers are more likely to churn.
- Higher monthly charges increase churn probability.
- Customers with longer tenure show lower churn risk.
- Electronic payment users have higher churn tendencies.

Future Improvements

- Hyperparameter tuning for better model performance.
- Use SMOTE for class imbalance handling.
- Deploy model using Streamlit or Flask.
- Integrate explainability using SHAP values.

Conclusion

- Machine Learning and Deep Learning can effectively predict telecom customer churn.
- The project helps businesses improve customer retention strategies.
- Predictive analytics can significantly reduce revenue loss.

References

- IBM Telco Customer Churn Dataset – Kaggle
- Scikit-learn Documentation
- TensorFlow / Keras Documentation
- XGBoost Documentation